

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07 11

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards *(BL2,BL 6)*

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters – more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable

fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a

teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiberbased ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This

hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
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Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach

Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Introduction

Network topology defines how computers, networking devices, and communication links are arranged within an organization. Selecting an appropriate topology depends on several factors such as number of users, cost, reliability, security, scalability, maintenance, bandwidth requirements, and fault tolerance.

Different organizations have different networking requirements. A small startup may prioritize simplicity and low maintenance, while a bank may require maximum redundancy and near-zero downtime. Similarly, a university campus requires a scalable architecture capable of supporting thousands of users across multiple buildings.

The following sections analyze five different industry scenarios and recommend suitable network topologies and

architectures.

1. STAR TOPOLOGY FOR A 20-PERSON STARTUP

1.1 Problem Statement

A startup has approximately 20 employees working in an office environment. The company requires a network that is:

- Easy to install

- Easy to manage

- Reliable

- Cost-effective

- Simple to troubleshoot

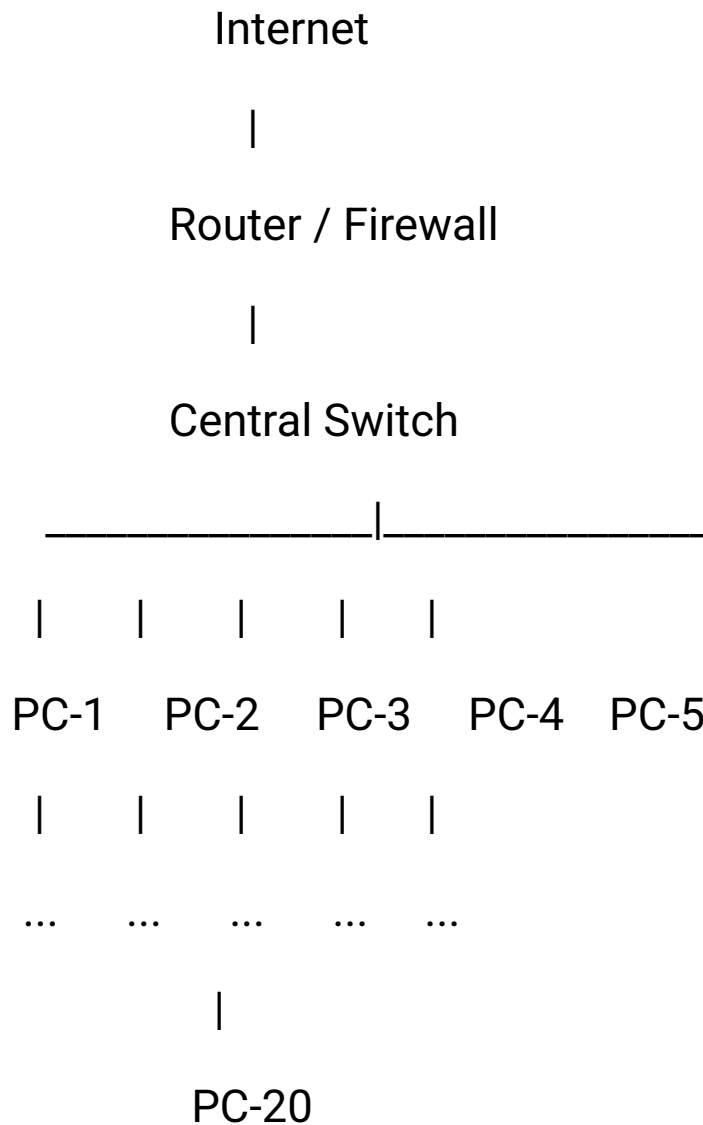
- Capable of future expansion

A star topology is an appropriate choice because every device connects to a central switch.

1.2 Proposed Network Topology

In a star topology, all computers and other network devices are connected to a central switch.

Basic Structure



The central switch becomes the main communication point for the office.

1.3 Recommended Hardware

1. Managed Switch

A 24-port Layer 2/Layer 3 managed switch is recommended.

Although there are only 20 employees, 24 ports provide

additional connections for:

- Printers

- VoIP phones

- Wireless access points

- Network storage

- Future employees

A managed switch also supports features such as:

- VLANs

- QoS

- Port monitoring

- Security policies

- Traffic management

2. Router/Firewall

A router connected to the switch provides internet connectivity.

The firewall protects the internal network from unauthorized external traffic.

The router/firewall can provide:

- Internet connectivity

DHCP

NAT

Firewall filtering

VPN connectivity

Routing

3. Cat6 UTP Cable

Each device can be connected to the switch using Cat6 UTP cables.

Cat6 supports Gigabit Ethernet over appropriate cable lengths, with a commonly specified Ethernet copper channel limit of up to 100 meters.

4. Wireless Access Point

A wireless access point connected to the switch provides connectivity for:

Laptops

Smartphones

Tablets

Wireless printers

1.4 IP Addressing

A private IP network can be created for the startup.

For example:

Network: 192.168.10.0/24

Gateway: 192.168.10.1

The router can assign IP addresses automatically using DHCP.

Example:

Device	IP Address
Router	192.168.10.1
PC-1	192.168.10.10
PC-2	192.168.10.11
PC-3	192.168.10.12
Printer	192.168.10.30

1.5 Working of the Network

When PC-1 wants to communicate with PC-2:

PC-1



Central Switch



PC-2

The switch examines the destination MAC address and forwards the frame through the appropriate port.

For internet access:

PC



Switch



Router/Firewall



Internet

1.6 Fault Isolation

One of the biggest advantages of star topology is fault isolation.

Suppose the cable connecting PC-5 to the switch fails.

PC-5 ---- X ---- Switch

Only PC-5 loses network connectivity.

The other users continue working:

PC-1 → Switch → Working

PC-2 → Switch → Working

PC-3 → Switch → Working

PC-4 → Switch → Working

PC-5 → Cable failure

Therefore, troubleshooting becomes easier.

1.7 Advantages

1. Easy installation.
2. Easy troubleshooting.
3. Failure of one cable affects only one device.
4. Easy to add new devices.

5. Centralized network management.
6. Good network performance.
7. Supports VLAN and security features.
8. Suitable for future expansion.

1.8 Limitations

The major limitation is that the central switch is a critical device.

If the switch fails:

All connected devices



Lose LAN connectivity

This can be reduced by using a higher-quality managed switch and maintaining a spare switch.

1.9 Why Star Topology Is Suitable

Requirement	Star Topology Solution
20 users	24-port switch

DOWNTIME

2.1 Problem Statement

A bank requires a highly reliable network because even a short interruption can affect:

- Customer transactions

- ATM services

- Online banking

- Internal applications

- Payment systems

- Financial databases

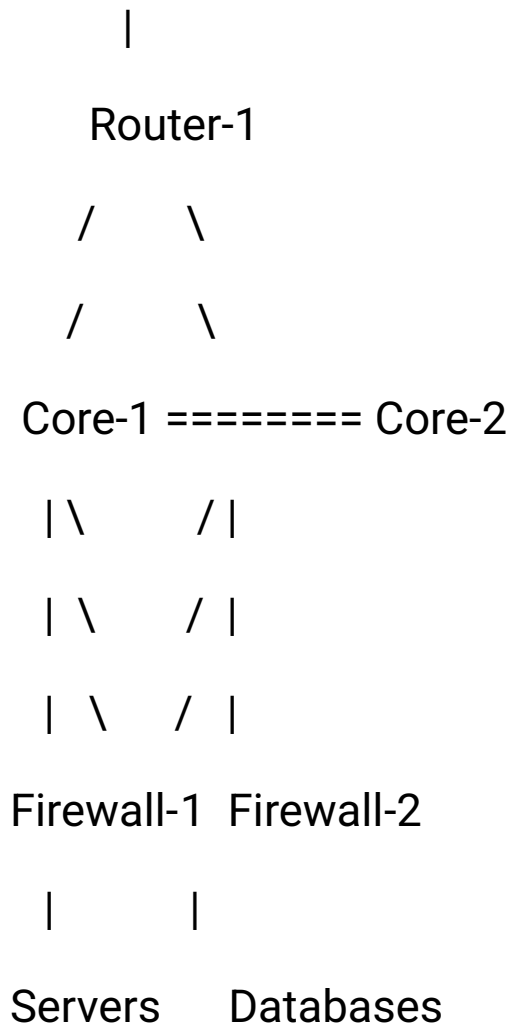
Therefore, the network must have high availability, redundancy, fault tolerance, and fast recovery.

A full or partial mesh topology is appropriate for the critical core network.

2.2 Proposed Topology

A practical banking network would normally use partial mesh redundancy rather than connecting every device directly to every other device.

Internet-1



Multiple links provide alternate paths.

2.3 Redundant Core Devices

The network should contain redundant:

Core switches

Routers

Firewalls

Internet connections

Power supplies

Critical servers

For example:

Primary Router

|

+----- Internet

|

Backup Router

|

+----- Backup Internet

If the primary router fails, traffic can be redirected through the backup router.

2.4 Fiber Optic Connectivity

Fiber optic cables are recommended for critical backbone links.

Single-Mode Fiber

Used for longer distances, such as connections between buildings.

Multimode Fiber

Suitable for shorter high-speed connections, such as within a data center where the deployment distance is appropriate.

Fiber provides:

- High bandwidth

- Low latency

- Long-distance capability

- Immunity to electromagnetic interference

2.5 Routing Redundancy

Dynamic routing protocols such as OSPF can be used within the organization.

For external connectivity, BGP may be used where appropriate.

Example:

Path 1:

Core → Router-1 → ISP-1

Path 2:

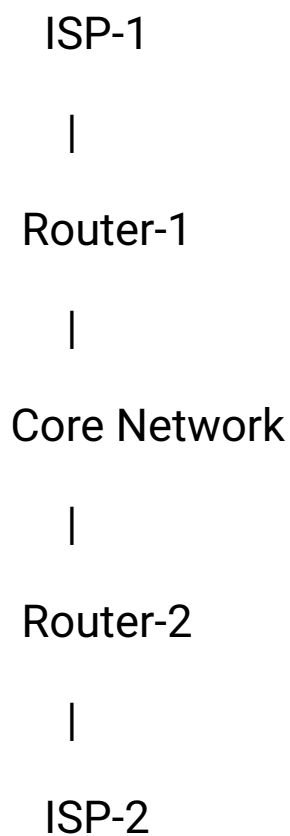
Core → Router-2 → ISP-2

If Path 1 becomes unavailable, routing can select Path 2.

2.6 Redundant Internet Connections

A bank should avoid depending on a single ISP.

A recommended arrangement is:



This provides connectivity even when one provider experiences an outage.

2.7 Redundant Power

Network redundancy alone is insufficient if all equipment depends on one power source.

The bank should therefore use:

- UPS systems

- Generator backup

- Dual power supplies

- Separate power circuits

2.8 Server and Storage Redundancy

Critical banking applications should also have redundancy.

Storage systems can use technologies such as RAID where appropriate.

Critical servers can be configured in high-availability clusters.

Therefore:

Primary Server

|

+---- Replication ----+

|

Backup Server

2.9 Failure Scenario

Suppose a link between Core-1 and Firewall-1 fails.

Because another path is available:

Core-1 → Core-2 → Firewall-2

traffic can continue through the alternate path.

This is the major benefit of mesh-based redundancy.

2.10 Advantages

1. High fault tolerance.
2. Multiple alternative paths.
3. Reduced downtime.
4. Improved availability.
5. Suitable for mission-critical banking systems.
6. Supports redundant internet connectivity.
7. Can automatically reroute traffic.
8. Protects critical services from individual link failures.

2.11 Limitations

1. Higher installation cost.
2. More cables and interfaces.
3. More complicated configuration.
4. Requires skilled network administrators.
5. More difficult monitoring and maintenance.

However, these additional costs are justified by the bank's requirement for high availability.

2.12 Why Mesh Is Suitable

Requirement	Mesh Solution
Near-zero downtime	Multiple paths
Link failure	Alternate path
Router failure	Redundant router
ISP failure	Multiple ISPs
High bandwidth	Fiber backbone
Power failure	UPS + generator

Data availability	Redundant servers/ storage
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Reliability	High redundancy
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2.13 Conclusion

A partial mesh topology with redundant core switches, routers, firewalls, fiber links, and internet connections is suitable for a banking environment. Although it requires greater investment and more complex management, the redundancy significantly improves availability and fault tolerance. For a bank, the additional cost is justified because continuous operation and reliable access to critical financial services are essential.

3. LOW-COST BUS TOPOLOGY FOR A CLASSROOM LAB

3.1 Problem Statement

A small classroom laboratory has a limited budget and only a small number of computers. The network is mainly used for:

- Internet browsing

File sharing

Basic educational activities

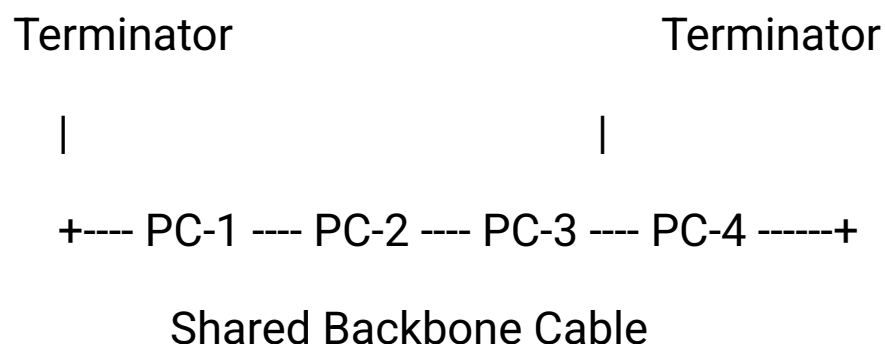
Sharing classroom resources

The network does not require highly advanced infrastructure.

A traditional bus topology is historically associated with low hardware requirements, although it is now largely obsolete in modern Ethernet LANs.

3.2 Traditional Bus Topology

In a traditional bus topology, all computers share one common backbone cable.



The cable carries communication signals between devices.

3.3 Components

Traditional coaxial Ethernet could use:

RG-58 coaxial cable

BNC connectors

T-connectors

50-ohm terminators

The terminators prevent signal reflections at the ends of the cable.

3.4 Working

When PC-1 transmits data:

PC-1

|

+-----+

|

Shared Bus |

|

PC-2 PC-3 PC-4

All devices can observe the transmission, but the intended destination processes the relevant frame.

3.5 Collision Problem

Because devices share the same communication medium, simultaneous transmissions can cause collisions in classic shared Ethernet.

For example:

PC-1 → Data

PC-3 → Data



Collision

CSMA/CD was historically used to manage such shared-medium Ethernet collisions.

3.6 Cable Failure

A major disadvantage is that failure of the backbone can affect multiple or all connected devices.

PC-1 ---- PC-2 ---- X ---- PC-3 ---- PC-4



Cable failure

This makes troubleshooting more difficult than in a star topology.

3.7 Low-Cost Alternative

For a modern classroom, an inexpensive unmanaged Ethernet switch with Cat5e/Cat6 patch cables is a more practical solution than constructing a traditional coaxial bus.

Unmanaged Switch

/ / | \ \

/ / | \ \

PC1 PC2 PC3 PC4 PC5

This is physically a star topology, but it retains the low-cost and simple-management objective.

3.8 Advantages

Traditional Bus

1. Low amount of backbone hardware.
2. Simple concept.
3. Lower initial cable/device requirements.

4. Suitable historically for small networks.

Modern Low-Cost Switch-Based Alternative

1. Easy installation.
2. Better performance.
3. Easier troubleshooting.
4. No shared coaxial backbone.
5. Affordable unmanaged switches are widely available.

3.9 Limitations

Traditional bus topology has significant limitations:

Backbone failure can disrupt many devices.

Shared medium can cause collisions.

Troubleshooting can be difficult.

Network expansion can be inconvenient.

Performance decreases as network traffic increases.

Traditional coaxial Ethernet is largely obsolete.

3.10 Why It May Be Considered for the Scenario

For a theoretical or legacy-network scenario with very

limited requirements, bus topology demonstrates how shared infrastructure can reduce hardware requirements.

However, for an actual modern classroom, a small unmanaged switch with Cat5e/Cat6 cabling is generally the more practical low-cost choice.

3.11 Conclusion

Traditional bus topology can minimize hardware requirements in a small, low-traffic environment, but it has important reliability and scalability limitations. Since classic coaxial Ethernet is no longer common in modern LAN deployments, an inexpensive unmanaged switch with Cat5e/Cat6 cabling is a more realistic modern solution. This provides better reliability and easier maintenance while remaining affordable.

4. HYBRID TOPOLOGY FOR A LARGE UNIVERSITY CAMPUS

4.1 Problem Statement

A large university has thousands of users distributed across:

- Lecture halls

- Laboratories

Libraries

Hostels

Administrative buildings

Research centers

Outdoor areas

A single topology cannot efficiently meet all these requirements.

Therefore, a hybrid topology is recommended.

4.2 Proposed Hybrid Architecture

The university can combine:

Mesh/ring redundancy at the backbone

Star topology inside buildings

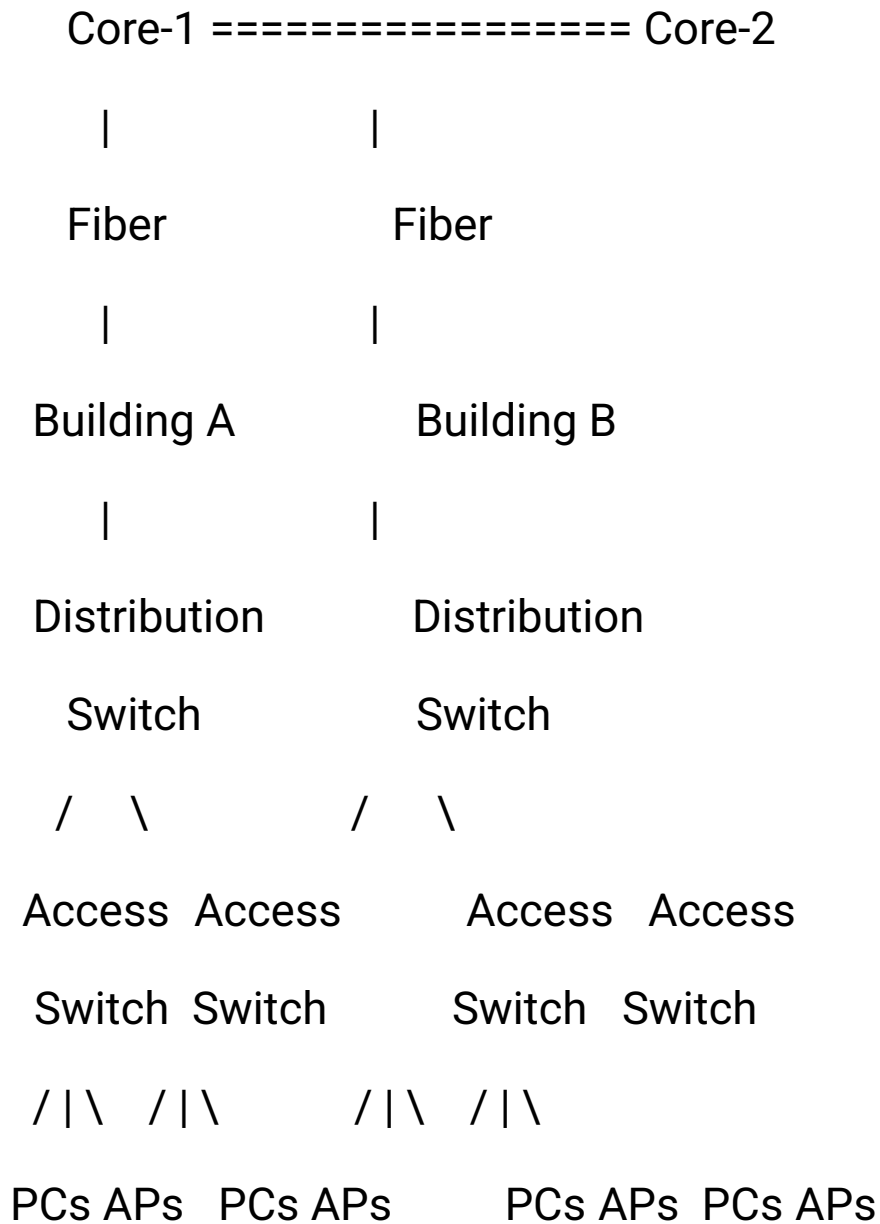
Hierarchical/tree structure for access and distribution

Wireless networks for mobile users

Example:

CAMPUS CORE



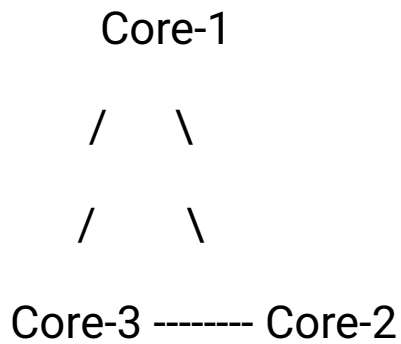


4.3 Campus Backbone

The campus core should use high-speed fiber links.

The backbone can use a ring or partial-mesh design for redundancy.

For example:



If one backbone link fails, traffic can use another path.

4.4 Building Distribution Layer

Each building should have one or more distribution switches.

Example:

Campus Core

|

Dual Fiber Uplinks

|

Distribution Switch

|

| | |

Access-1 Access-2 Access-3

Dual uplinks improve reliability.

4.5 Access Layer

Access switches connect end-user devices.

Devices include:

- Desktop computers

- Printers

- IP phones

- Wireless access points

- Laboratory systems

Cat6 copper cabling can be used for appropriate horizontal connections.

4.6 Wireless Network

Wireless access points can be deployed throughout:

- Classrooms

- Libraries

- Hostels

- Cafeterias

Common areas

PoE-capable access switches can supply power and network connectivity to compatible APs.

Access Switch

|

PoE

|

Wireless AP

/ | \

Laptop Phone Tablet

4.7 VLAN Segmentation

Different groups should be logically separated.

Example:

VLAN Users

VLAN
10 Students

VLAN
20 Faculty

VLAN	Administrati
30	on

VLAN	Guests
40	

VLAN	Laboratories
50	

VLAN	Servers
60	

VLANs reduce unnecessary broadcast traffic and improve security.

4.8 Three-Layer Architecture

A campus network can follow a hierarchical model.

Core Layer

Provides:

- High-speed routing

- Backbone connectivity

- Redundancy

Distribution Layer

Provides:

- Policy enforcement

- Routing between VLANs

- Aggregation of access switches

Access Layer

Provides:

- End-device connectivity

- User access

- PoE for APs and IP phones

4.9 Scalability

Suppose the university constructs a new building.

The existing network does not need to be completely redesigned.

The new building can be connected as:

New Building

|

Distribution Switch

|

Dual Fiber

|

Campus Core

This makes expansion easier.

4.10 Fault Tolerance

Suppose one fiber link fails:

Building A

|

X Fiber Failure

|

Campus Core

If a second fiber uplink exists:

Building A

/ \

/ \

Fiber Fiber

| |

Core-1 Core-2

communication can continue through the alternate connection.

4.11 Advantages

1. Highly scalable.
2. Suitable for thousands of users.
3. Supports multiple buildings.
4. Provides backbone redundancy.
5. Easy building-level management.
6. Supports wired and wireless devices.
7. Allows VLAN segmentation.
8. Simplifies future expansion.
9. Supports high-speed fiber backbone.
10. Provides better fault isolation.

4.12 Limitations

1. Higher initial cost.
2. Requires skilled administrators.
3. More networking equipment.

4. More complex configuration.
5. Requires proper network planning.

4.13 Why Hybrid Topology Is Suitable

Requirement	Solution
Thousands of users	Hierarchical architecture
Multiple buildings	Fiber backbone
Reliability	Ring/partial mesh
Building management	Star topology
Wireless users	Wi-Fi APs
Security	VLAN segmentation
Expansion	Modular building design
High bandwidth	Fiber backbone

4.14 Conclusion

A hybrid topology is the most appropriate solution for a large university campus because different parts of the campus have different requirements. A redundant fiber backbone provides reliability, star-based access networks provide easy management, and hierarchical design provides scalability. Wireless access points provide mobility, while VLANs provide logical separation and security. This combination creates a flexible and scalable campus network.

5. HIERARCHICAL NETWORK FOR A MULTI-DEPARTMENT COMPANY

5.1 Problem Statement

A company has several departments such as:

Human Resources

Finance

Sales

IT

Administration

The organization requires a network that provides:

Efficient communication

Security

Departmental separation

Internet connectivity

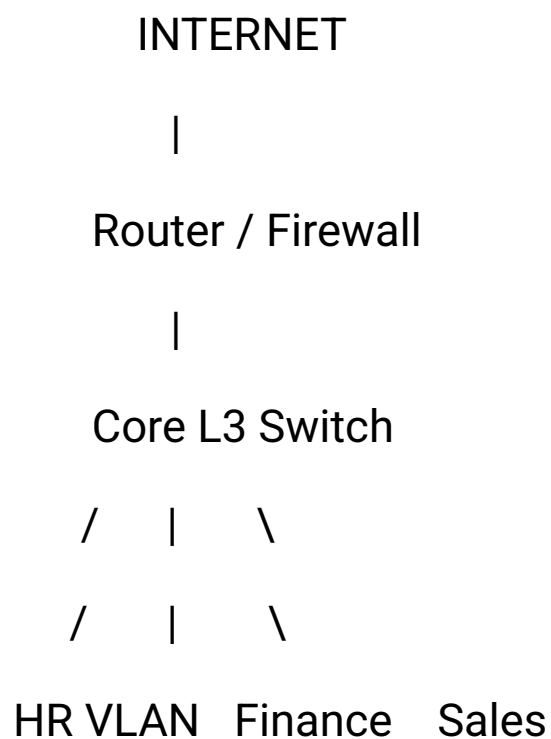
Easy management

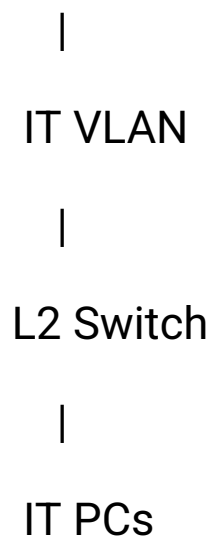
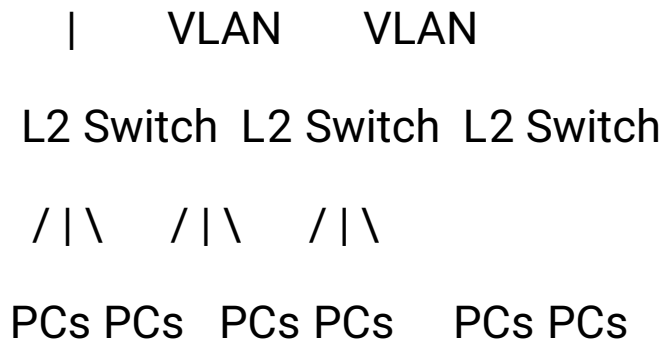
Scalability

Reliability

A hierarchical network architecture using VLANs, Layer 2 switches, and Layer 3 routing is recommended.

5.2 Proposed Network Architecture





5.3 Access Layer

Each department can have a dedicated Layer 2 access switch.

For example:

HR

|

HR Switch

|

Computers + Printer

Finance

|

Finance Switch

|

Computers + Printer

This allows devices within each department to communicate efficiently.

5.4 VLAN Configuration

VLANs logically separate departments even when they share the same physical switching infrastructure.

Example:

Department	VLAN
	N
HR	10
Finance	20

Sales 30

IT 40

Administrati
on 50

Guest 60

5.5 Why VLANs Are Important

Without VLANs, all departments could potentially share the same broadcast domain.

With VLANs:

VLAN 10 → HR

VLAN 20 → Finance

VLAN 30 → Sales

VLAN 40 → IT

This provides:

- Logical separation

- Reduced broadcast domains

- Better security

Easier management

5.6 Inter-VLAN Routing

Different departments may need to communicate.

For example:

HR VLAN

|

↓

Layer 3 Switch

|

↓

Finance VLAN

A Layer 3 switch can perform inter-VLAN routing.

Access control rules can determine which departments are allowed to communicate.

For example:

Sales → HR

Allowed only for specific services

Sales → Finance

Restricted

IT → All departments

Administrative access according to policy

5.7 Router and Firewall

The router/firewall connects the internal company network to the internet.

Internal Network

|

L3 Switch

|

Router/Firewall

|

Internet

The firewall can control:

Incoming traffic

Outgoing traffic

VPN connections

Web access

Unauthorized connections

5.8 IP Addressing and Subnetting

Each VLAN can have a separate subnet.

Example:

VLAN	Network
HR	192.168.10.0 /24
Finance	192.168.20.0 /24
Sales	192.168.30.0 /24
IT	192.168.40.0 /24
Administration	192.168.50.0 /24

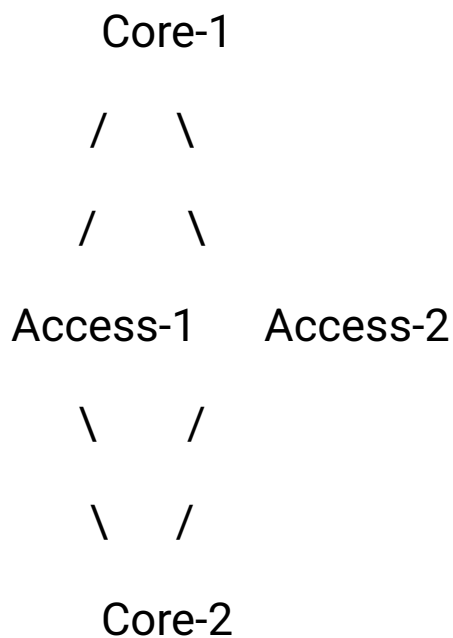
The default gateway for each VLAN can be configured on

the Layer 3 switch.

5.9 Redundancy

For improved reliability, critical links can be duplicated.

Example:



If one uplink fails, the second link can continue carrying traffic when appropriately configured.

Redundant core devices can also be used in larger organizations.

5.10 Security

Security can be implemented at multiple levels.

VLAN Security

Separates departments.

Firewall

Protects the organization from external threats.

Access Control

Controls which users and departments can access sensitive resources.

Authentication

Users should use individual accounts and strong authentication.

Port Security

Switch ports can be configured to restrict unauthorized devices where appropriate.

5.11 Quality of Service

Certain applications may require priority.

For example:

VoIP Traffic → High Priority

Business Applications → Medium/High Priority

Normal Web Traffic → Normal Priority

Guest Traffic → Lower Priority

QoS can improve the performance of voice and other delay-sensitive applications.

5.12 Advantages

1. Better security.
2. Reduced broadcast traffic.
3. Easy department management.
4. Easy expansion.
5. Improved network performance.
6. Supports inter-department communication.
7. Supports internet access.
8. Allows access-control policies.
9. Supports redundancy.
10. Suitable for growing companies.

5.13 Example of Communication

Suppose an employee in Sales wants to access an

approved Finance application.

The communication follows:

Sales PC



Sales Access Switch



Layer 3 Switch



Firewall / ACL Policy



Finance Application

The network administrator can control this communication using routing and security policies.

5.14 Scalability

If a new department is created, for example Research and Development, a new VLAN and subnet can be created.

R&D VLAN = 70

Network = 192.168.70.0/24

A new access switch can then be connected to the existing network.

Therefore, the company can expand without redesigning the entire infrastructure.

5.15 Requirement Analysis

Requirement	Proposed Solution
Department separation	VLANs
Local communication	Layer 2 switches
Inter-department communication	Layer 3 routing
Internet access	Router
Security	Firewall + ACLs
IP management	Subnetting + DHCP
Reliability	Redundant links
Performance	Managed switches + QoS
Future expansion	Hierarchical

architecture

Wireless access

Access points

5.16 Conclusion

A hierarchical network using Layer 2 access switches, VLANs, a Layer 3 core/distribution switch, and a router/firewall is highly suitable for a multi-department company. VLANs provide logical separation and reduce broadcast traffic, while inter-VLAN routing allows controlled communication between departments. Security policies, firewall protection, proper IP addressing, and redundant links improve reliability and security.

The architecture is also scalable because new departments, users, switches, and VLANs can be added without redesigning the entire network.

OVERALL COMPARISON OF THE FIVE INDUSTRY PROBLEMS

Scenario	Recommended Topology	Main Reason
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20-Person Startup	Star	Simple, manageable and reliable
Multi- Department Company	Hierarchical / Extended Star with VLANs	Security + performance + scalability

FINAL CONCLUSION

The choice of network topology should always be based on the organization's requirements and constraints rather than selecting the same topology for every situation.

For a small startup, star topology is preferred because it is simple, easy to troubleshoot, and scalable. A bank requires mesh-based redundancy because continuous availability and fault tolerance are critical. A small classroom with a very limited budget may be discussed using a traditional bus topology, although modern Ethernet networks generally use inexpensive switches instead. A university campus requires a hybrid topology because it combines redundant backbone connectivity with manageable building-level networks. Finally, a company with multiple departments benefits from a hierarchical architecture with VLANs and Layer 3 routing, providing security, performance, and

scalability.

Therefore, effective network design is a constraint-based decision-making process in which topology, hardware, cabling, security, redundancy, cost, and future growth are considered together to produce a reliable and efficient network.